HARARE INSTITUTE OF TECHNOLOGY

SCHOOL OF INORMATION SCIENCE & TECHNOLOGY

B(TECH) HONS DEGREE IN SOFTWARE ENGINEERING

PART 2 SEMESTER 1

ISE 2103: DATA COMMUNICATIONS AND NETWORKS

TIME: 3 HOURS **DATE: NOV/DEC 2024**

TOTAL MARKS: 100

INSTRUCTIONS TO CANDIDATES:

- *This question paper contains a total of **FIVE** questions.*
- *Each question carries **25** marks*
- *Answer **any four questions of your choice.***
- *Start a new question on a fresh page*

ADDITIONAL MATERIAL:

- *None.*

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QUESTION 1

a) Explain the following terms:
   i. Half Duplex Transmission. [1]
   ii. Point to Point communication. [1]
   iii. Synchronous transmission. [1]

b) Describe the two components that make up a multiplexor. [4]
   i. Outline the difference between Bounded and unbounded transmission media. [2]
   ii. Explain the functional difference between a Repeater and Bridge. [2]
   iii. Compare and contrast Global system for mobile communication (GSM) and Code division multiple access (CDMA) wireless technology. [3]
   iv. State the difference between Dynamic and Static IP address. [2]

c) Explain two WAN protocols. [2]

d) Signals picks unwanted random energy, discuss such three transmission impairments. [3]

e) Discuss the significance of TCP/IP and its implementation in data networks. [4]

QUESTION 2

a) Differentiate between Frequency Division Multiplexing (FDM) and Time Division multiplexing (TDM) techniques. [5]

b) In the description of the Ethernet Architecture, explain the function and operation of the carrier Sense Multiple Access with Collision Detection Flow control paradigm. [5]

c) Describe the following switching techniques as used in data communication and networking:
   i. Circuit switching. [2]
   ii. Packet Switching. [2]

d) Discuss any four network security threats you are familiar with. [3]

e) Explain the limitations of a ring network topology compared to a bus network topology. [3]

f) With the aid of diagram, describe the complete computer communication systems. [5]

QUESTION 3

a) Explain any three error detection strategy used to control data transmission. [3]

b) With the aid of well labelled diagram of OSI model, describe the functions provided by each layer. [9]

c) Distinguish between the following terms.
   i. Logical verses Physical topology. [2]
   ii. Analogue verses digital signals. [2]

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d) Describe in detail four possible configurations of computer networks as categorized by relationships among nodes. **[4]**

e) Identify appropriate network topology and network connecting devices for following requirements. Draw a network design for proposed network for the following scenario. An organization having its office in a building of 5 floors. Each floor needs 20 machines. **[5]**

QUESTION 4

a) Distinguish between the internet and the World Wide Web. **[4]**

b) State the four benefits of client –server networks compared to peer to peer networks. **[4]**

c) Using a diagram, explain the features of the types of networking media used in enterprise networks. **[4]**

d) A basic computer network must consist of a least hardware, software and protocols. Give examples of each of these networking requirements and explain their implementation. **[4]**

e) Write brief notes about the competencies expected of the following professionals.

i. A network administrator. **[1]**

ii. A network Engineer. **[2]**

iii. A hacker. **[1]**

f) Differentiate between Hub & switch on the basis of layer, ports, device type, speed. **[5]**

QUESTION 5

a) Explain the harm caused by the following.

i. Trojans. **[1]**

ii. Worms. **[1]**

iii. Ransomware. **[1]**

iv. Spam. **[1]**

b) Distinguish between modulation techniques verses encoding. Give two examples of each. **[6]**

c) The internet offers a variety of services to user. List and explain any four. **[4]**

d) With reference to the client-server paradigm, explain how Harare Institute of Technology students are able to access online learning resources remotely from home. **[7]**

e) Discus the differences between TCP and UDP. **[4]**

END OF EXAMINATION PAPER

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